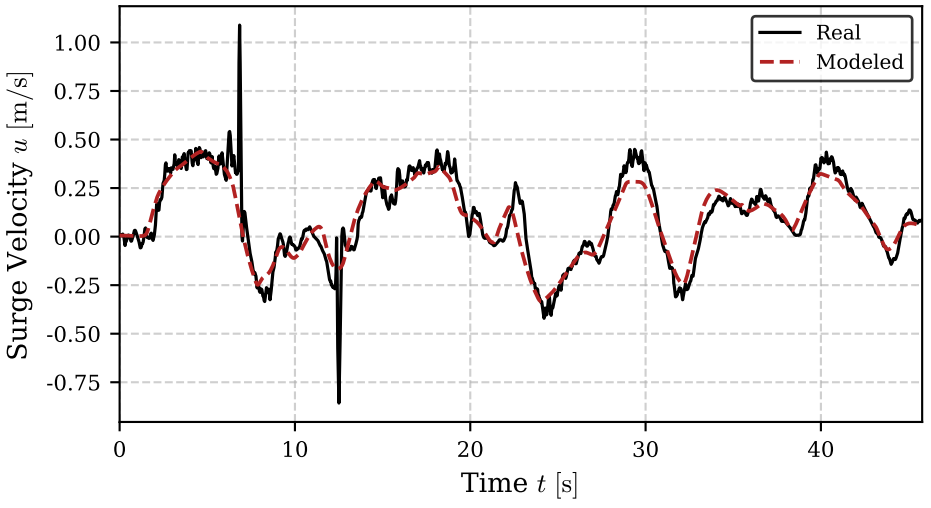
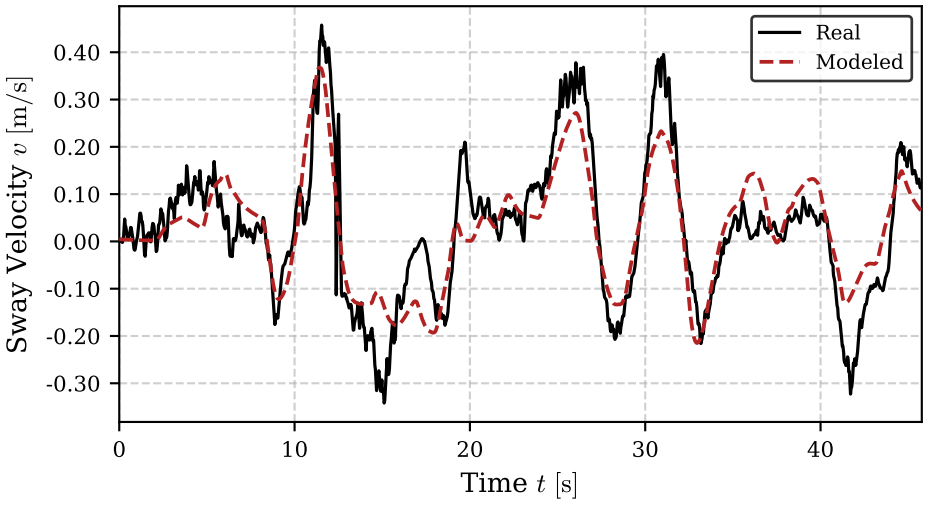
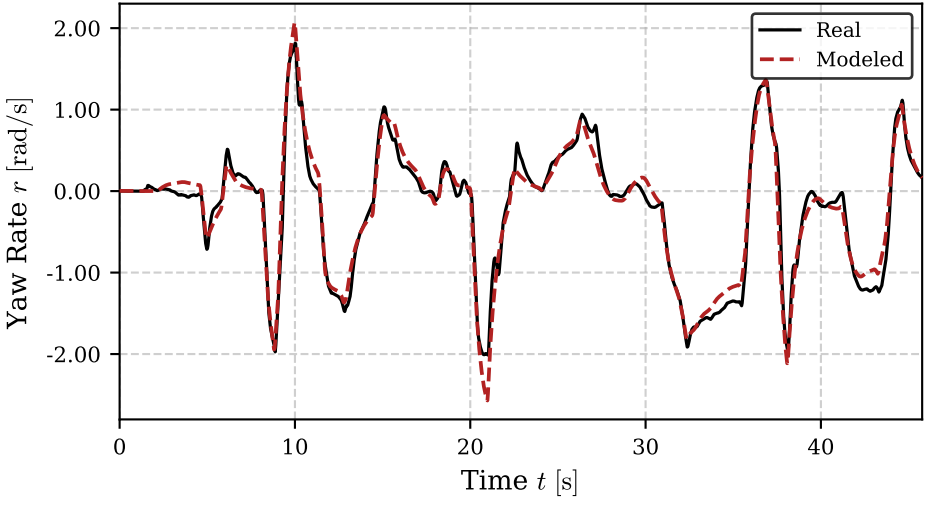
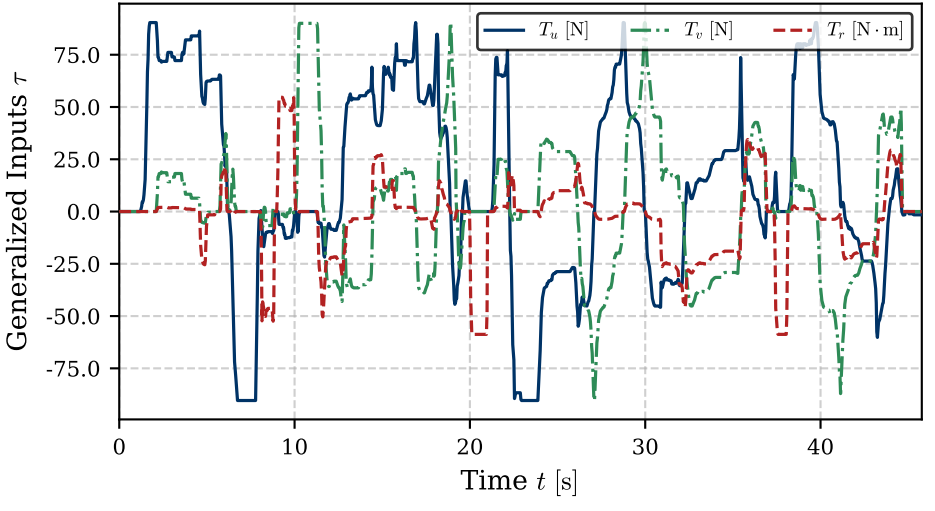
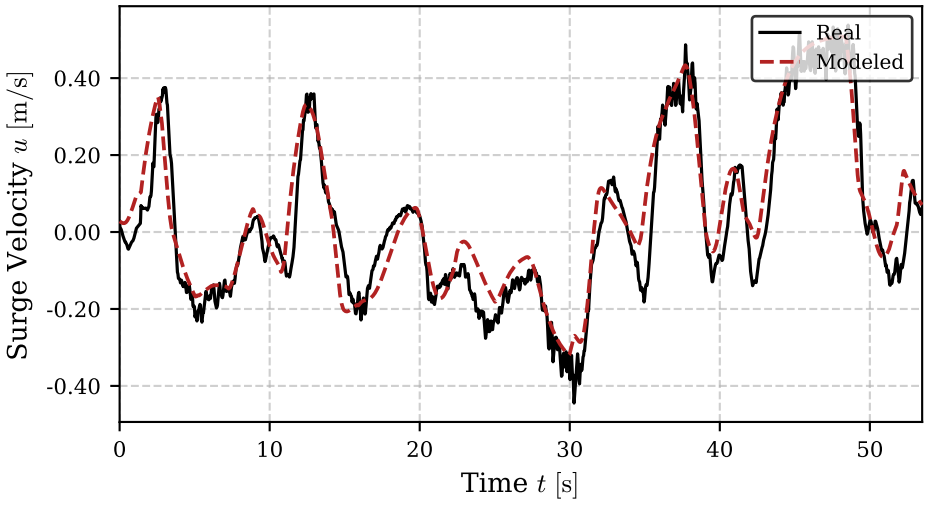
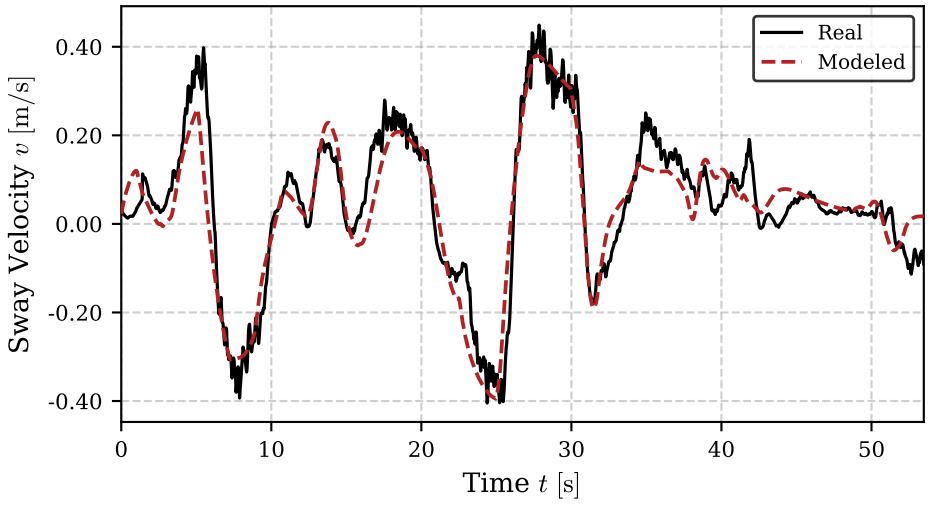

Real
Modeled
Surge Velocity $u$ [m/s]
Time $t$ [s]

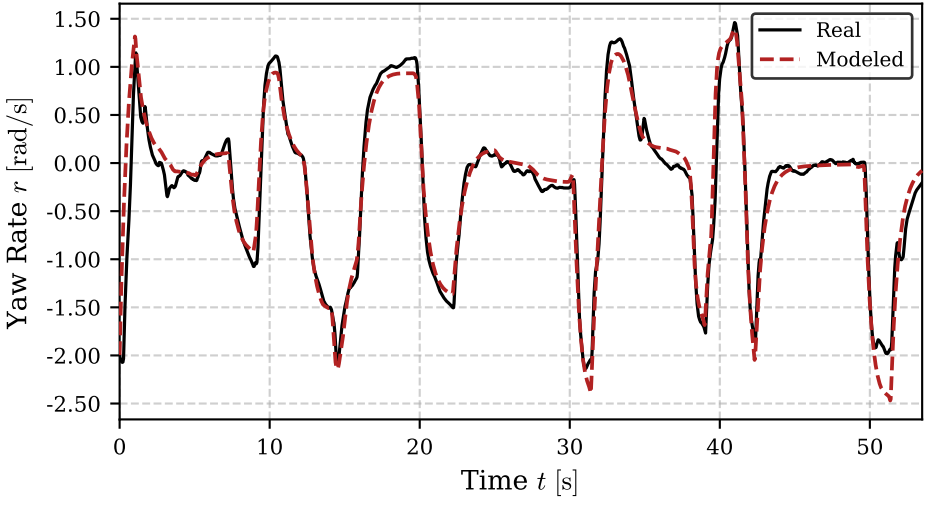

Real
Modeled
Sway Velocity $v$ [m/s]
Time $t$ [s]

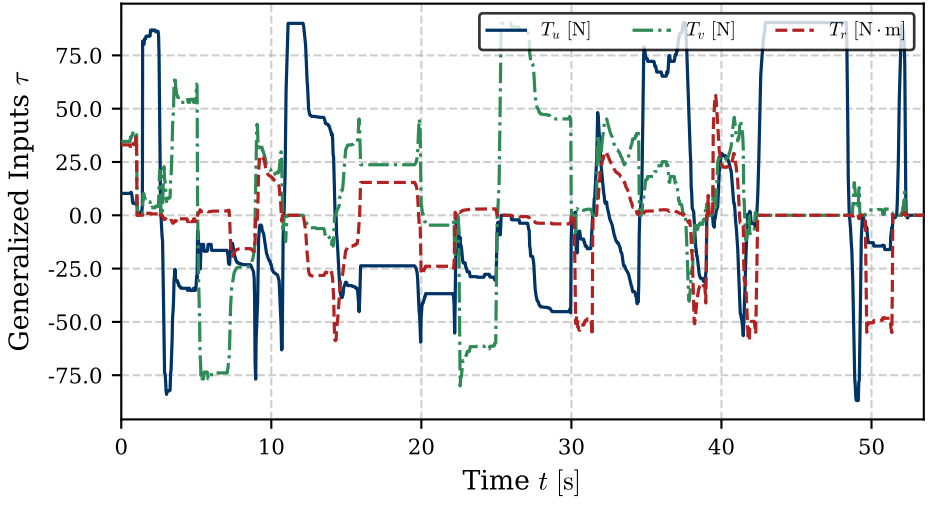

Real
Modeled
Yaw Rate $r$ [rad/s]
Time $t$ [s]

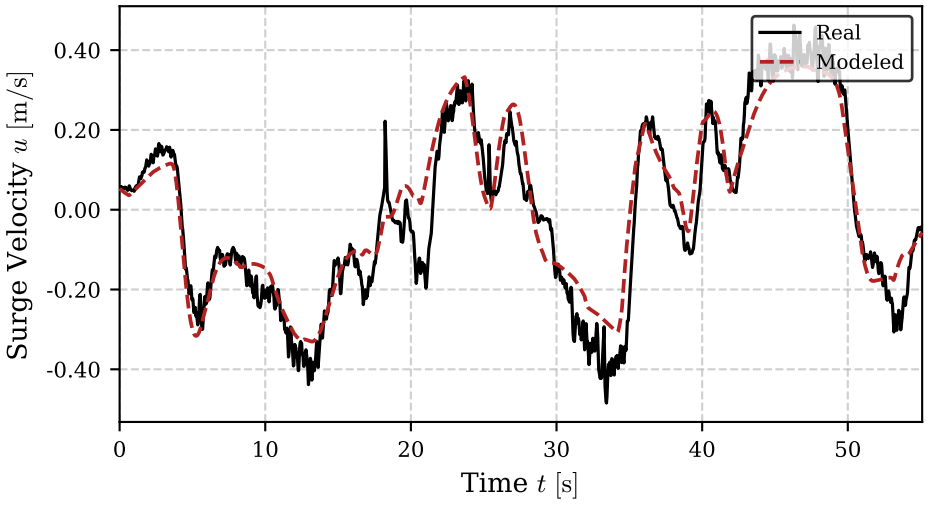

$T_u$ [N]
$T_v$ [N]
$T_r$ [N · m]
Generalized Inputs $\tau$
Time $t$ [s]

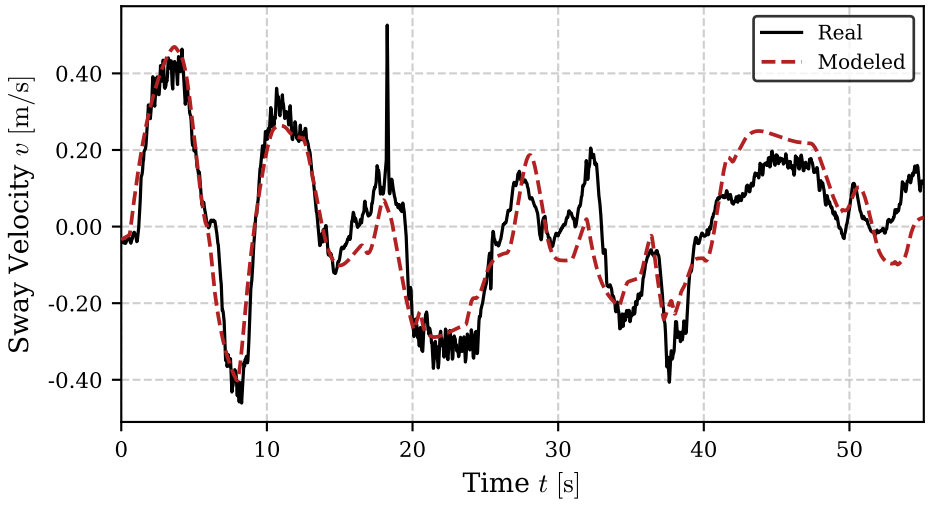

Real
Modeled
Surge Velocity $u$ [m/s]
0.40
0.20
0.00
-0.20
-0.40
0
10
20
30
40
50
Time $t$ [s]

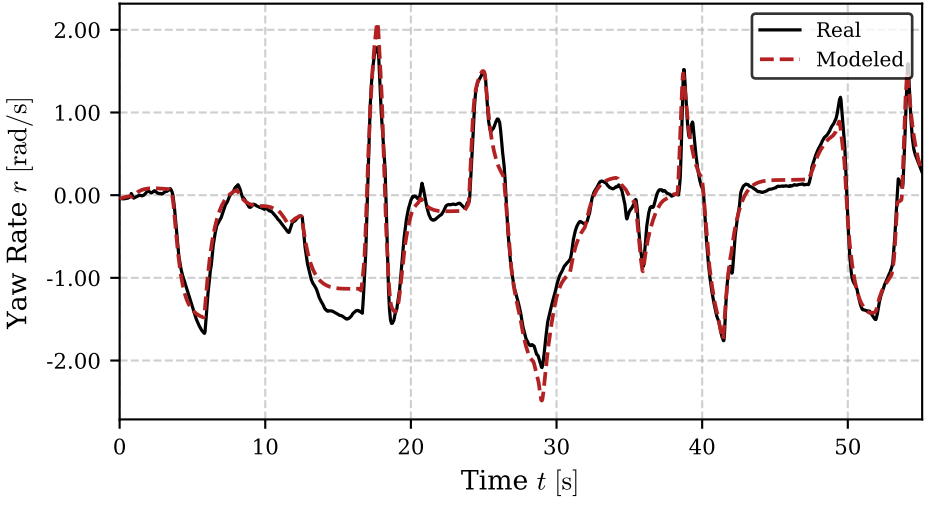

Real
Modeled
Sway Velocity $v$ [m/s]
0.40
0.20
0.00
-0.20
-0.40
0
10
20
30
40
50
Time $t$ [s]

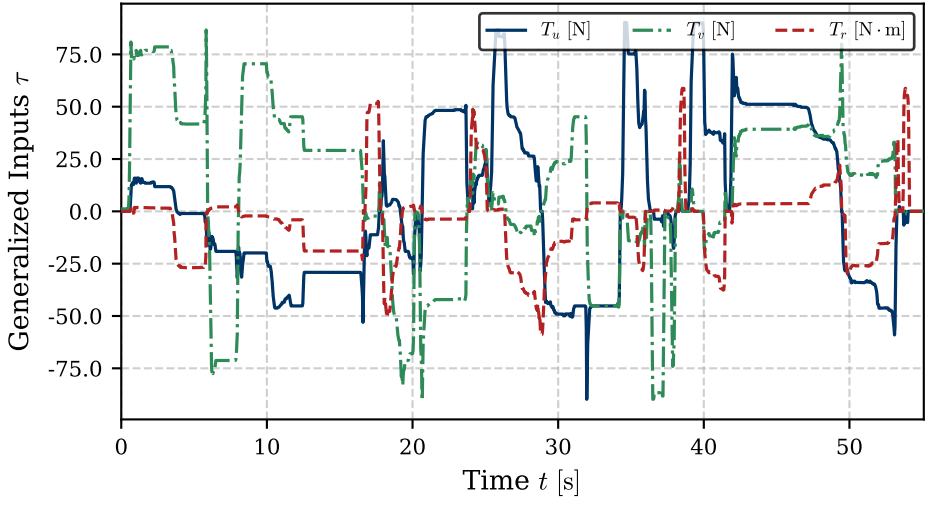

Real
Modeled
Yaw Rate $r$ [rad/s]
1.50
1.00
0.50
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
0
10
20
30
40
50
Time $t$ [s]

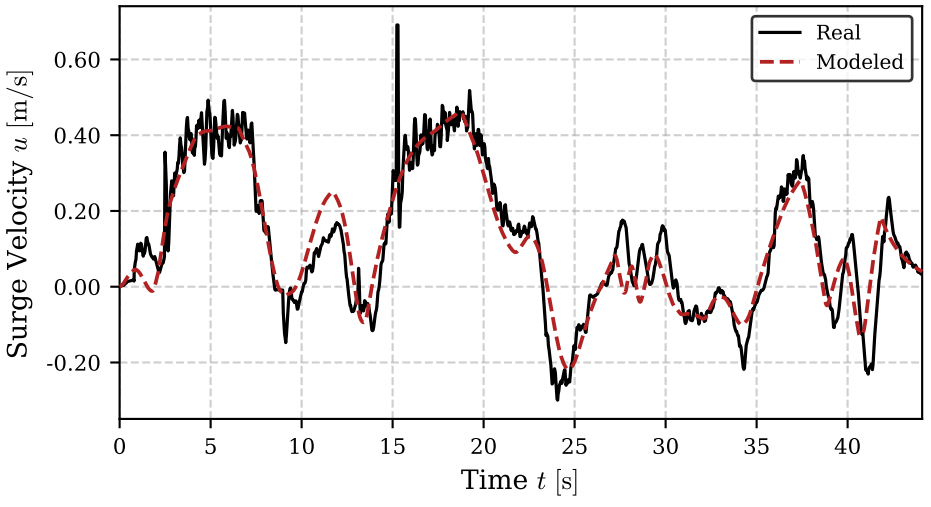

$T_u$ [N]
$T_v$ [N]
$T_r$ [N · m]
Generalized Inputs $\tau$
75.0
50.0
25.0
0.0
-25.0
-50.0
-75.0
0
10
20
30
40
50
Time $t$ [s]

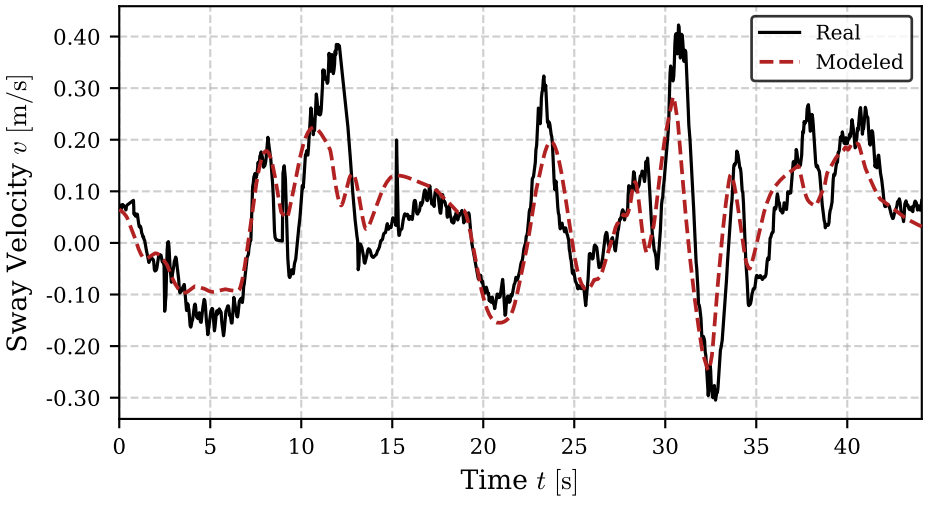

Real
Modeled
Surge Velocity $u$ [m/s]
0.40
0.20
0.00
-0.20
-0.40
0
10
20
30
40
50
Time $t$ [s]

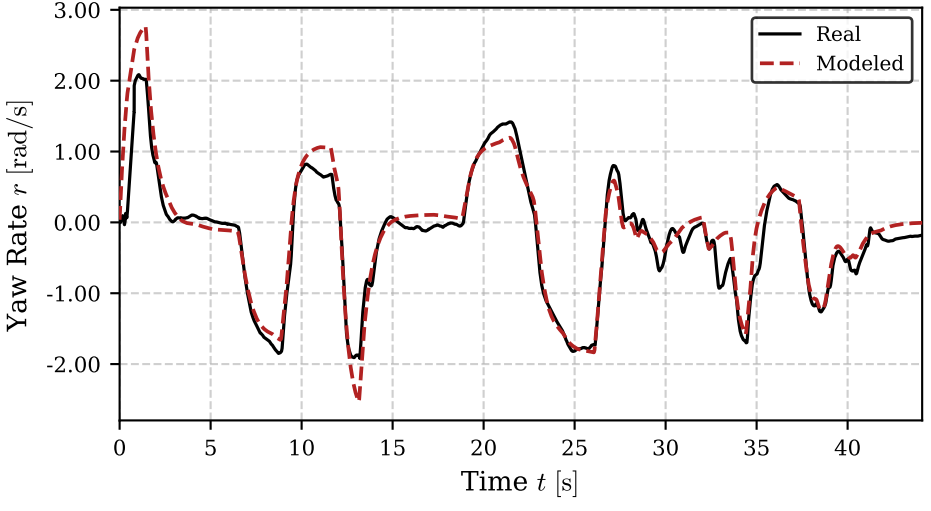

Real
Modeled
Sway Velocity $v$ [m/s]
0.40
0.20
0.00
-0.20
-0.40
0
10
20
30
40
50
Time $t$ [s]

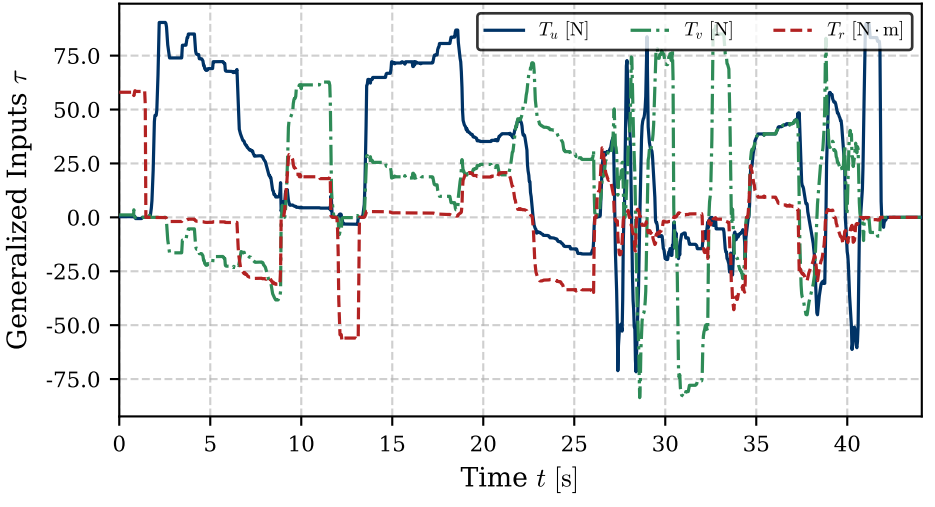

Real
Modeled
Yaw Rate $r$ [rad/s]
2.00
1.00
0.00
-1.00
-2.00
0
10
20
30
40
50
Time $t$ [s]

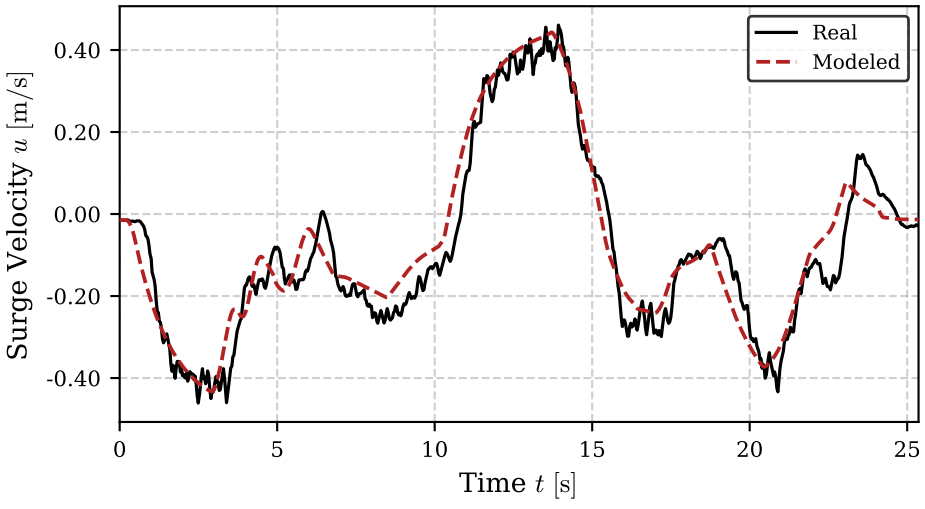

$T_u$ [N]
$T_v$ [N]
$T_r$ [N · m]
Generalized Inputs $\tau$
75.0
50.0
25.0
0.0
-25.0
-50.0
-75.0
0
10
20
30
40
50
Time $t$ [s]

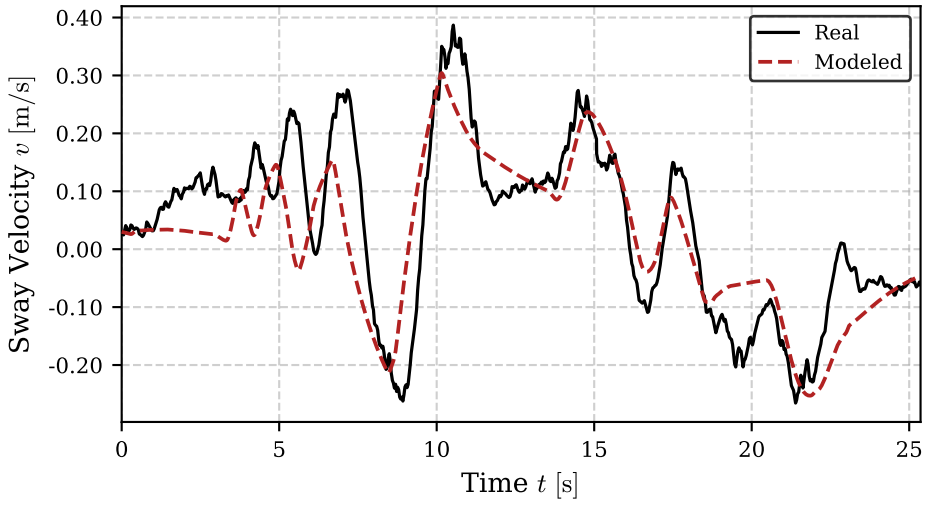

Real
Modeled
Surge Velocity $u$ [m/s]
Time $t$ [s]

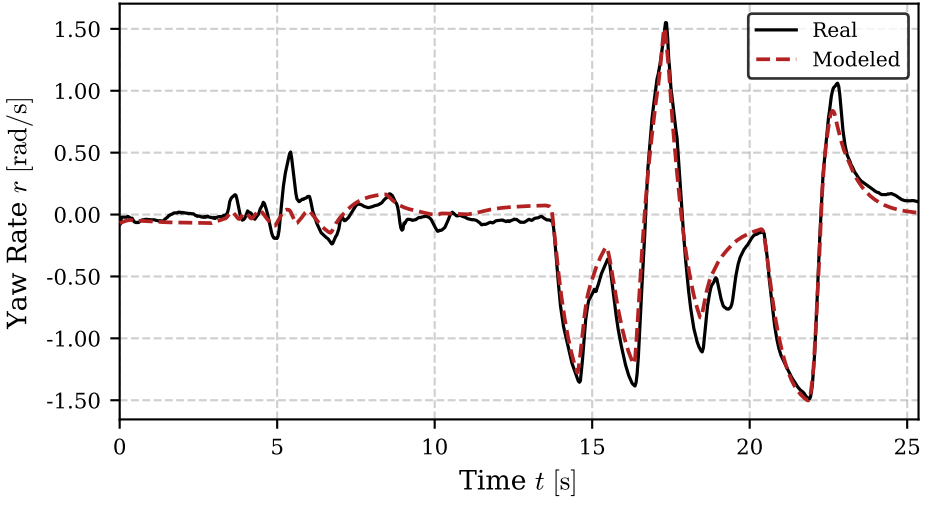

Real
Modeled
Sway Velocity $v$ [m/s]
Time $t$ [s]

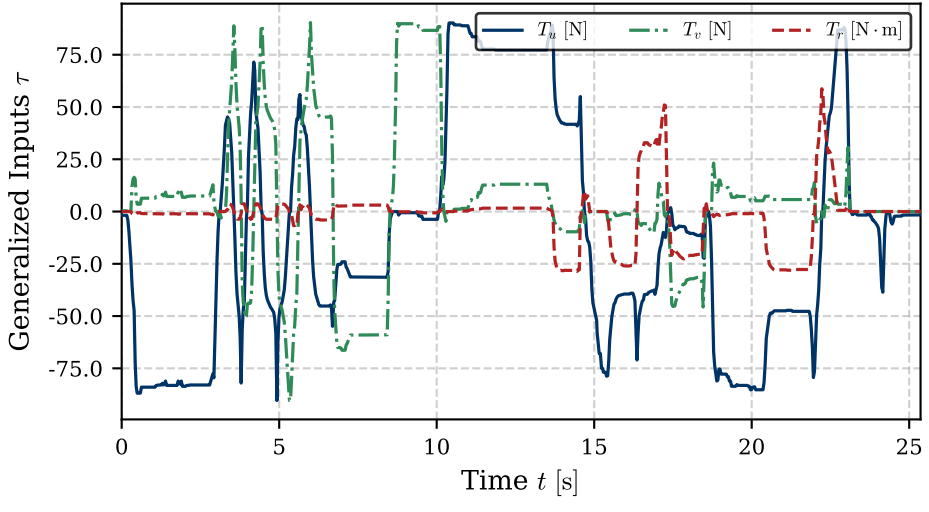

Real
Modeled
Yaw Rate $r$ [rad/s]
Time $t$ [s]

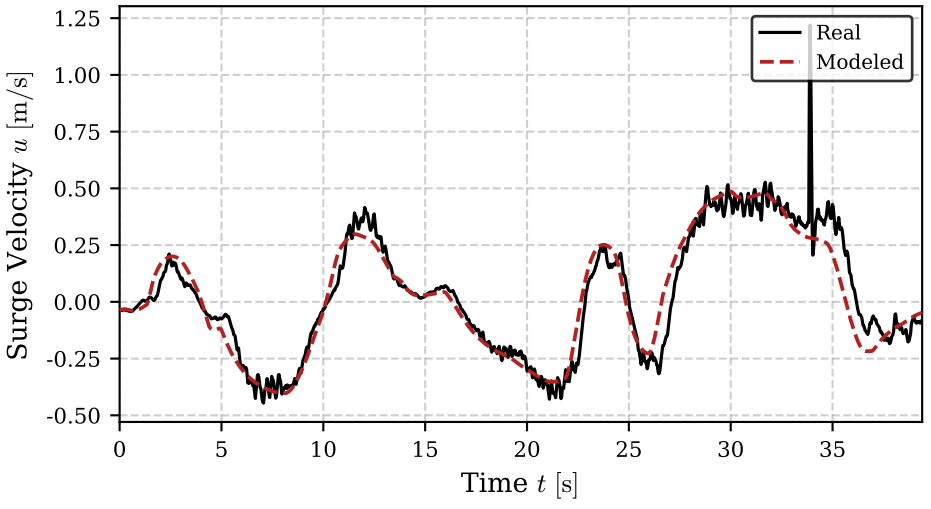

$T_u$ [N]
$T_v$ [N]
$T_r$ [N · m]
Generalized Inputs $\tau$
Time $t$ [s]

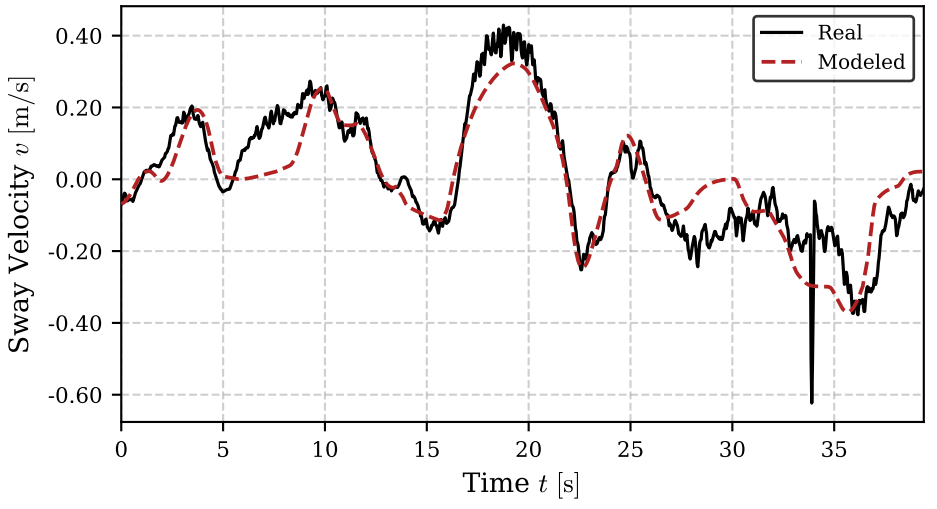

Real
Modeled
Surge Velocity $u$ [m/s]
Time $t$ [s]

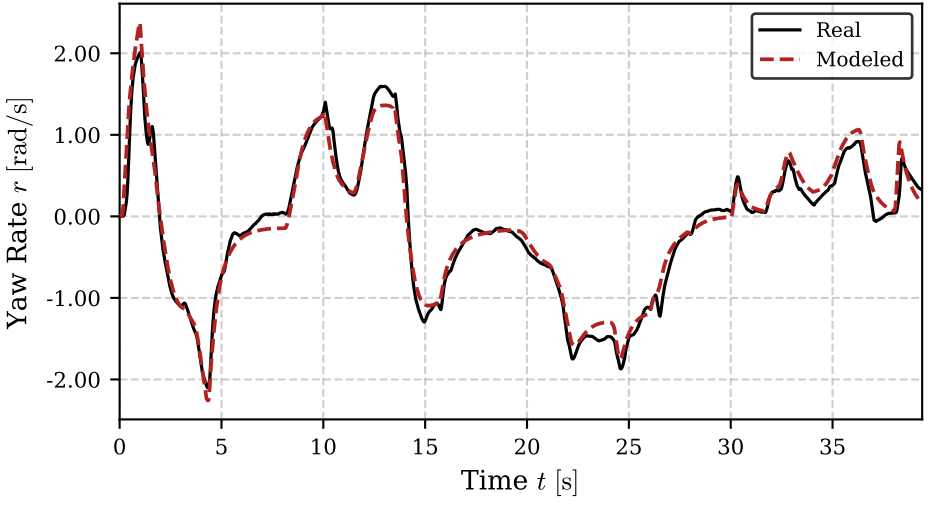

Real
Modeled
Sway Velocity $v$ [m/s]
Time $t$ [s]

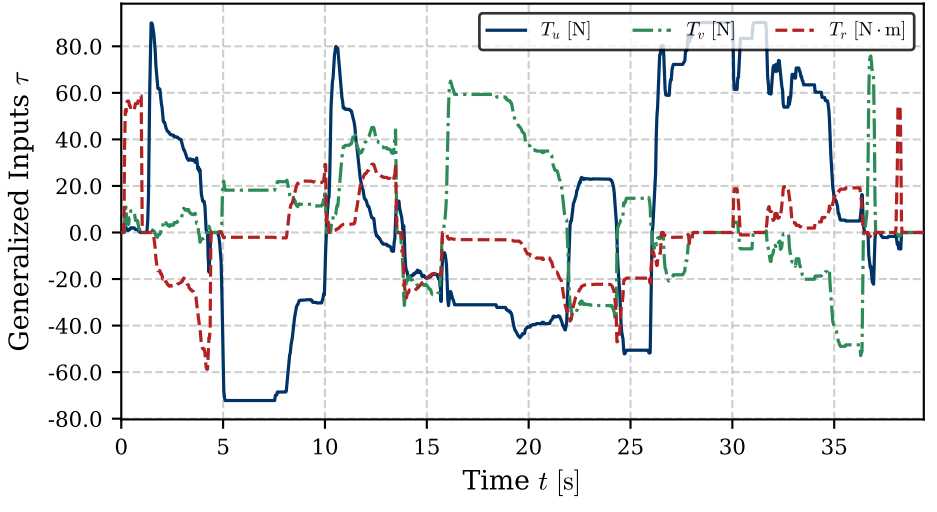

Real
Modeled
Yaw Rate $r$ [rad/s]
Time $t$ [s]


$T_u$ [N]
$T_v$ [N]
$T_r$ [N · m]
Generalized Inputs $\tau$
Time $t$ [s]


Real
Modeled
Surge Velocity $u$ [m/s]
Time $t$ [s]


Real
Modeled
Sway Velocity $v$ [m/s]
Time $t$ [s]


Real
Modeled
Yaw Rate $r$ [rad/s]
Time $t$ [s]


$T_u$ [N]
$T_v$ [N]
$T_r$ [N · m]
Generalized Inputs $\tau$
Time $t$ [s]